Extensions to Hierarchical Clustering

- Weakness of the agglomerative & divisive hierarchical clustering methods
  - No revisit: cannot undo any merge/split decisions made before
  - Scalability bottleneck: Each merge/split needs to examine many possible options
    - Time complexity: at least $O(n^2)$, where n is the number of total objects
- Several other hierarchical clustering algorithms
  - BIRCH (1996): Use CF-tree and incrementally adjust the quality of sub-clusters
  - CURE (1998): Represent a cluster using a set of well-scattered representative points
  - CHAMELEON (1999): Use graph partitioning methods on the K-nearest neighbor graph of the data

BIRCH: A Multi-Phase Hierarchical Clustering Method

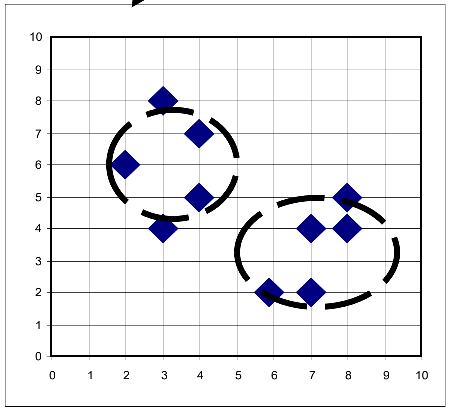
- BIRCH (Balanced Iterative Reducing and Clustering Using Hierarchies)
  - Developed by Zhang, Ramakrishnan & Livny (SIGMOD'96)
  - Impact many new clustering methods and applications (received 2006 SIGMOD Test of Time award)
- Major innovation
  - Integrating hierarchical clustering (initial micro-clustering phase) and other clustering methods (at the later macro-clustering phase)
- Multi-phase hierarchical clustering
  - Phase1 (initial micro-clustering): Scan DB to build an initial CF tree, a multi-level compression of the data to preserve the inherent clustering structure of the data
  - Phase 2 (later macro-clustering): Use an arbitrary clustering algorithm (e.g., iterative partitioning) to cluster flexibly the leaf nodes of the CF-tree

Clustering Feature Vector

- Consider a cluster of multi-dimensional data objects/points
- The clustering feature (CF) of the cluster is a 3-D vector summarizing info about clusters of objects
  - Register the 0-th, 1st, and 2nd moments of a cluster
- Clustering Feature (CF): CF = <n, LS, SS>
  - n: Number of data points
  - LS: linear sum of n points: $\boldsymbol{LS} = \sum_{i=1}^{n} \boldsymbol{x}_i$:
  - SS: square sum of n points: $SS = \sum_{i=1}^{n} \|\boldsymbol{x}_i\|^2$

n = 5; **LS** = ((3+2+4+4+3), (4+6+5+7+8)) = (16, 30);
SS=($3^2+2^2+4^2+4^2+3^2$) + ($4^2+6^2+5^2+7^2+8^2$)= 54+190=244

CF_1 = <5, (16,30), 244>



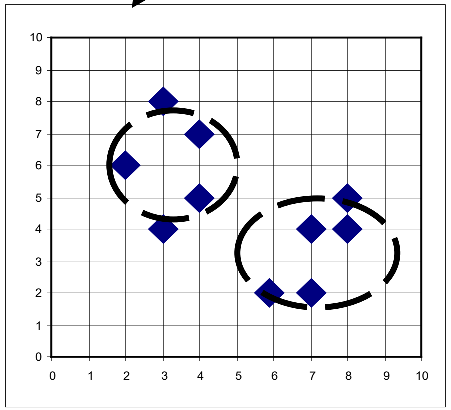
(3,4)
(2,6)
(4,5)
(4,7)
(3,8)

Clustering Feature: a Summary of the Statistics for the Given Cluster

- Registers crucial measurements for computing cluster and utilizes storage efficiently
- Clustering features are additive: Merging clusters C1 and C2—linear summation of CFs

$$CF_1 + CF_2 = \langle n_1 + n_2, LS_1 + LS_2, SS_1 + SS_2 \rangle$$

CF_1 = <5, (16,30), 244>

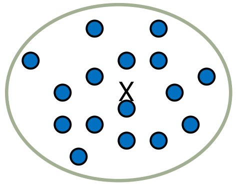


(3,4)
(2,6)
(4,5)
(4,7)
(3,8)

n = 5; **LS** = ((3+2+4+4+3), (4+6+5+7+8)) = (16, 30);
SS=$(3^2+2^2+4^2+4^2+3^2)$ + $(4^2+6^2+5^2+7^2+8^2)$= 54+190=244

Essential Measures of Cluster: Centroid, Radius and Diameter

- Centroid: x_0
  - The "middle" of a cluster
  - n: number of points in a cluster
  - x_i is the i-th point in the cluster



$$\boldsymbol{x}_0 = \frac{\sum_{i=1}^{n} \boldsymbol{x}_i}{n} = \frac{\boldsymbol{LS}}{n}$$

$$R = \sqrt{\frac{\sum_{i=1}^{n} \|\boldsymbol{x}_i - \boldsymbol{x}_0\|^2}{n}} = \sqrt{\frac{SS}{n} - \frac{\|\boldsymbol{LS}\|^2}{n^2}}$$

$$D = \sqrt{\frac{\sum_{i=1}^{n} \sum_{j=1}^{n} \|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2}{n(n-1)}} = \sqrt{\frac{2n \cdot SS - 2\|\boldsymbol{LS}\|^2}{n(n-1)}}$$

- Radius: R
  - Average distance from member objects to the centroid
  - The square root of average distance from any point of the cluster to its centroid
- Diameter: D
  - Average pairwise distance within a cluster
  - The square root of average mean squared distance between all pairs of points in the cluster

Example

- Continuing the previous example:
- $CF = < 5, (16, 30), 244 >$
- $\|\boldsymbol{LS}\|^2 = 16^2 + 30^2 = 1156$
- $R = \sqrt{244/5 - 1156/5^2} = 1.60$
- $D = \sqrt{(2 \times 5 \times 244 - 2 \times 1156)/(5 \times 4)} = 2.53$